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$La_2Ce_2O_7$ supported ruthenium as a robust catalyst for ammonia synthesis☆

Wenfeng Han*, Zhi Li, Huazhang Liu

Institute of Industrial Catalysis, Zhejiang University of Technology, Hangzhou 310032, China

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ABSTRACT

$La_2Ce_2O_7$ nanoparticles were prepared by citric acid complexation method followed by calcination at varied temperatures. Then, supported with 4 wt% Ru, they were evaluated as the catalysts for ammonia synthesis under conditions similar with industry. With $La_2Ce_2O_7$ being calcinated at 700 or 800 °C, the experimental results indicate that the Ru/$La_2Ce_2O_7$ catalyst exhibits much higher ammonia concentration or ammonia synthesis rate than that of Ru/CeO_2 and Ru/La_2O_3. In addition, Ru/$La_2Ce_2O_7$ possesses high stability under over-heating test. In the absence of any promotor, ammonia concentration of Ru/$La_2Ce_2O_7$ catalyst approaches 14% at 450 °C, GHSV of 10000 h^{-1} and pressure of 10 MPa. The rate-determining step of ammonia synthesis, dissociation of N_2 is significantly facilitated by the strong metal–support interaction (SMSI) between Ru and $La_2Ce_2O_7$. Due to the interaction, $La_2Ce_2O_7$ tends to donate electrons to Ru, resulting in the high electron density over the surface of Ru active sites which is favorable for the dissociation of N_2. Consequently, high activity is achieved.

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1. Introduction

Ammonia is the feedstock for chemical fertilizer production, and it is the key nitrogen source for other chemicals.[1] In addition, as a CO_2 and CO free source, ammonia is considered as a potential and promising carrier of H_2 energy.[2] Generally, it is well accepted that N_2 dissociation is the rate-determining step (RDS) during ammonia synthesis.[3] As the N≡N bond (945 kJ/mol) is too strong to be cleaved, the high temperature (400–600 °C) and high pressure (20–40 MPa) are usually adopted in industrial ammonia synthesis,[4] which leads to high energy consumption. Ammonia synthesis process is one of the main energy consumers, accounting for about 1% of global energy consumption.[5] During ammonia synthesis, catalyst plays an important role in the cost and energy efficiency.[6] Consequently, the development of a proper catalyst is highly desired.

Following the exploration of suitable ammonia synthesis catalyst for more than 100 years,[7] Ru-based catalysts are considered as the second-generation ammonia synthesis catalysts with high activity under mild conditions after well consolidated iron catalyst.[8,9] To obtain higher activity for ammonia synthesis, various supports, such as MgO, Al_2O_3, $BaTiO_3$ and BN were evaluated as the supports of Ru catalysts.[10–13] Among various kinds of supports, the catalysts with Ru supported on activated carbon or similar carbon materials (promoted with Ba, K or Cs) display relatively high activity.[14–16] Unfortunately, Ru also catalyzes the methanation of carbon support under ammonia synthesis conditions, leading to the degradation of the support.[17–19] Hence, more recently, significant effort is devoted to the development of novel supports.[20–22]

It was reported that rare earths are potential support that can reduce the inhibition of H_2 adsorption on the dissociation of N_2 (H_2 strongly competes with N_2 adsorption over active sites).[23] In addition, promoted with lanthanide oxides, Ru atoms are negatively charged.[23,24] As a result, the energy level of the molecular orbital of N_2 adsorbed on the surface of Ru is altered. Therefore, in this respect, rare earth oxides are potentially effective support for ammonia synthesis.

In recent years, $La_2Ce_2O_7$ was confirmed to be a promising proton-electron conductor material for the applications such as solid oxide fuel cells (SOFCs), polymer proton exchange membrane, hydrogen separation.[25–27] Calcium doped $La_2Ce_2O_7$ proton conductor has also been investigated as the electrolyte for ammonia synthesis, and the maximum rate of ammonia synthesis is 1.3×10^{-9} mol/(cm^2·s) at atmospheric pressure, which is comparable to the capability of proton conduction in state-of-art of perovskite proton conductor.[28] The electron effect of support is the

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* Corresponding author.
E-mail address: hanwf@zjut.edu.cn (W.F. Han).

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most important factor of activating N_2 adsorbed on the surface of Ru. Hence, $La_2Ce_2O_7$ can be the suitable support for ammonia synthesis catalyst.[29] Indeed, it has been confirmed that Ru/$La_{0.5}Ce_{0.5}O_{1.75}$ pre-reduced at an unusually high temperature (650 °C) shows extremely high rates under mild conditions.[21,30] Although the catalyst was evaluated at much lower pressure compared with industry, it shows unexpected high stability and activity.

In the present study, $La_2Ce_2O_7$ nanoparticle was prepared by citric acid complexation method followed with calcination at varied temperatures. Then, it was evaluated as the support of Ru catalysts for ammonia synthesis at 450 °C, GHSV of 10000 h^{-1} and pressure of 10 MPa, close to industrial conditions. The crystal structure, morphology, surface area, porosity, surface properties and dispersion of Ru were investigated following the calcination of $La_2Ce_2O_7$ support at 500–900 °C.

2. Experimental

2.1. Preparation of catalysts

First, La_2O_3 (AR, Aladdin, China) was converted to nitrate via the reaction with HNO_3 solution. Then, desired amounts of $(NH_4)_2Ce(NO_3)_6$ (AR, Aladdin, China) was added to the above nitrate solution under vigorous stirring. To form the La and Ce complex, solid citric acid with molar ratio of citric acid/(cerium + lanthanum) = 2:1 was added to the solution. Followed by slow evaporation of the solution at 80 °C until a viscous liquid was obtained, the gel was dried at 110 °C. Finally, the solid sample was calcined in a muffle furnace at temperatures of 500, 600, 700, 800, 900 and 1000 °C for 5 h, respectively. Consequently, white power $La_2Ce_2O_7$ supports were obtained and they are denoted as $La_2Ce_2O_7$-500, $La_2Ce_2O_7$-600, $La_2Ce_2O_7$-700, $La_2Ce_2O_7$-800, $La_2Ce_2O_7$-900 and $La_2Ce_2O_7$-1000, respectively.

$Ru_3(CO)_{12}$ was prepared by the reaction of $RuCl_3 \cdot xH_2O$ (Aladdin, Ru content of 35.0 wt%–42.0 wt%) according to the procedure of our previous study.[14] Ruthenium catalyst was prepared by impregnation method with $La_2Ce_2O_7$ as the support and $Ru_3(CO)_{12}$ dissolved in THF solution as the precursor. The nominal Ru loadings are 4 wt% for all the catalysts. After 4 h, the solid paste was filtered out from the solution followed by drying in air at 110 °C for 5 h. Then, the powders were pressed as pellets at about 20 MPa. With crushing and sieving, catalysts with the sizes between 1.0 and 1.4 mm were collected (2 mL) for the activity test.

2.2. Evaluation of catalysis performance

Ammonia synthesis reactions were carried out in a four-tube fixed-bed integral reactor at the temperatures of 375, 400, 425 and 450 °C, space velocity of 10000 h^{-1}, and pressure of 3, 5 and 10 MPa. The catalyst bed was loaded in the isothermal zone of the reactor, and a thermocouple was placed at the isothermal zone of each tube reactor. Hydrogen and nitrogen mixture with a stoichiometric ratio of 3:1 was derived from ammonia decomposition (NH_3 gas purity 99.99%) and was further purified by Pd molecular sieve catalyst, 13X and 5A molecular sieve. After purification, the gas was pressurized by membrane compressor, and then passed through the reactors. Prior to ammonia synthesis, the catalysts were activated by the mixture of N_2 and H_2 (H_2:N_2 = 3:1) at a pressure of 5.0 MPa with a space velocity of 30000 h^{-1} at 400, 425, 450 and 475 °C for 4, 8, 8 and 4 h, respectively. Over-heating tests were carried out at a pressure of 5.0 MPa, space velocity of 30000 h^{-1} and at 500 °C for 12 h. Ammonia concentration was determined by titration with a sulfuric acid solution.[14]

2.3. Characterization of supports and catalysts

The physical adsorption experiment of N_2 at −196 °C was carried out on the Autosorb gas adsorption analyzer. With BET and BJH (Barret Joyer and Halenda) equations, the specific surface area, pore volume and pore size were determined. Prior to the measurement, samples were treated at 250 °C for 3 h.

The phase structure and crystal size of the catalysts were determined by the Swiss Thermo ARL X'TRA X-ray diffractometer (XRD). The voltage was 45 kV, and the current was 40 mA with a scanning speed of 0.20 (°)/min.

The morphology of the catalysts was investigated by a Hitachi S-4700 scanning electron microscope (SEM) with a working voltage of 15 kV. It is also equipped with an X-ray energy spectrometer (EDS) for the determination of elemental content. Transmission electron microscopy (TEM) were adopted to further explore the microstructure of catalysts on a Tecnai G2 F30 S-TWIN instrument from Philips-FEI Company working at 200 at 300 kV.

For X-ray photoelectron spectroscopy (XPS) experiments, a spectrometer from Kratos AXIS Ultra DLD photoelectron spectroscopy with a monochromatized microfocused Al X-ray source was employed. Charging of samples was corrected by setting the binding energy of adventitious carbon (C 1s) at 284.6 eV. Prior to the measurements, the powder sample was pressed into self-supporting disks and loaded into a subchamber. After evacuated at 25 °C for 4 h, the spectra were taken.

CO chemisorption was carried out on an AutoChem II 2920 analyzer. The sample (200 mg) was pre-reduced in a flow of pure H_2 (30 mL/min) at 400 °C for 2 h and then purged by He flow for 0.5 h. After cooled to room temperature, CO was injected at regular intervals until the area of the recorded peaks became constant. Ru dispersion and particle size were obtained from the cumulative volume of CO, assuming a chemisorption stoichiometry CO/Ru of 1:1.

3. Results

3.1. Synthesis and characterization of $La_2Ce_2O_7$ and catalysts

$La_2Ce_2O_7$ nanoparticles were prepared by citric acid complexation method followed with calcination at varied temperatures. The samples were first investigated by XRD and the results are presented in Fig. 1(a). Clearly, the formation of $La_2Ce_2O_7$ solid solution is identified following calcination at 500 °C, which is significantly lower than 600 °C reported in literature.[31] Hence, it indicates that citric acid complexation plays an important role in the formation of $La_2Ce_2O_7$ solid solution. With the increase in calcination temperature, the diffraction peaks of $La_2Ce_2O_7$ become sharper, and the half-maximum width becomes narrower. Compared with the CeO_2 spectrum, significant shift towards small angle is observed following the formation of solid solution. By contrast, no clear peaks of La_2O_3 are found in $La_2Ce_2O_7$. It is suggested that La^{3+} enters the lattice of CeO_2 crystalline and results in the increase in the lattice constant of $La_2Ce_2O_7$.[32] The disappearance of La_2O_3 diffraction peaks further indicates the formation of $La_2Ce_2O_7$ and La_2O_3 is dispersed in the solid solution completely. Therefore, the crystal facets of CeO_2 were used to represent the crystal planes of $La_2Ce_2O_7$.[31]

As no noticeable changes was detected following the impregnation of Ru, XRD spectra of Ru supported catalysts are not included in Fig. 1(b). For comparison, only fresh Ru/$La_2Ce_2O_7$-600 are included in Fig. 1(b), indicating that the Ru impregnation does not change the crystal structure of $La_2Ce_2O_7$. Following ammonia synthesis reaction at 450 °C for 10 h, almost identical XRD patterns to corresponding $La_2Ce_2O_7$ supports were recorded over spent

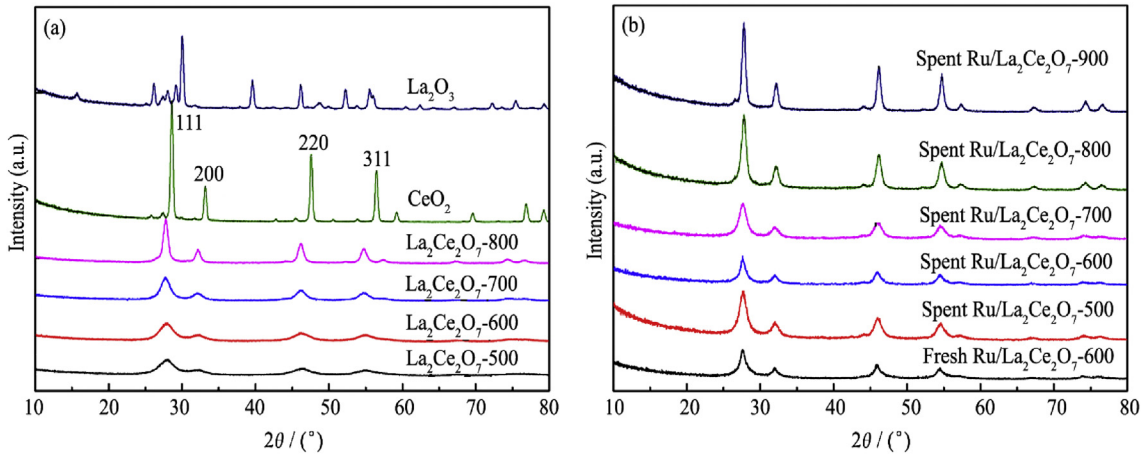

Fig. 1. XRD patterns for supports and catalysts. (a) La_2O_3, CeO_2 and $La_2Ce_2O_7$ after calcination at various temperatures; (b) Spent Ru/$La_2Ce_2O_7$ with $La_2Ce_2O_7$ supports calcinated at various temperatures.

catalysts (Fig. 1(b)). In addition, as displayed in Fig. 1(b), no diffraction peaks of Ru are identified on the spent catalysts even after ammonia synthesis at 450 °C. It suggests that Ru is well dispersed on the $La_2Ce_2O_7$ support.

The crystal sizes of all the planes derived from Scherrer equation for $La_2Ce_2O_7$ calcinated at different temperatures are listed in Table 1. As comparison, spent Ru/$La_2Ce_2O_7$ catalysts are also included.

As demonstrated in Table 1, no clear difference in crystal size is observed between $La_2Ce_2O_7$ supports and their corresponding spent catalysts. Since the supports were calcinated at temperatures above 500 °C, no sintering of support is found after ammonia synthesis reaction. However, during the preparation of $La_2Ce_2O_7$ supports, most of crystal sizes increase with calcination temperature. Furthermore, with calcination temperature increasing from 500 to 700 °C, only slight increase in crystal size is detected. Especially, a clear decrease in the crystal size of (311) facet is noticed. Consequently, as catalyst supports, it is expected to affect the catalytic performance significantly.

The morphology of catalysts was further characterized by SEM and the images are illustrated in Fig. 2. Following calcination of $La_2Ce_2O_7$ at 500 °C, aggregates of small particles are observed (Fig. 2(a)). Clearly, pore structure resulted from the aggregation of small particles. With calcination temperature of $La_2Ce_2O_7$ increasing from 500 to 700 °C, no significant change in morphology is identified (Fig. 2(b)). However, at calcination temperature of 800 °C, slight sintering is found as some of belt-like structures are found (Fig. 2(c)). At 900 °C, $La_2Ce_2O_7$ seems to sinter into large and solid particles (Fig. 2(d)). As suggested by the SEM images, a calcination temperature higher than 800 °C is not favorable as sintering is detected at this temperature, which agrees well with the results of XRD.

Fig. 3 illustrates the typical distribution of Ce, La and Ru over Ru/$La_2Ce_2O_7$-800 catalyst as determined by EDX. Clearly, catalyst is composed by Ce and La with Ru uniformly dispersed over the surface. As indicated in Fig. 3, the EDX map of Ce is almost identical with that of La, indicating the formation $La_2Ce_2O_7$ solid solution. Although the support was calcinated at 800 °C before the impregnation of Ru, relatively high dispersion is achieved.

Ru/$La_2Ce_2O_7$ catalysts were further investigated by TEM technique and the images are displayed in Fig. 4. Consistent with SEM results, the catalysts are in the form of small particles. With the $La_2Ce_2O_7$ calcination temperature increased from 500 to 900 °C, clear increase in the particle size is detected. Unexpectedly, although with high magnitude during the measurement of TEM, no clear Ru particles were observed supported on $La_2Ce_2O_7$-500, $La_2Ce_2O_7$-700 and $La_2Ce_2O_7$-800 (Fig. 4(a), (b) and (c)), respectively. By contrast, with $La_2Ce_2O_7$ calcinated at 900 °C, uniform Ru particles were detected with the sizes between 1 and 2 nm (Fig. 4(d)). Clearly, Ru is dispersed over $La_2Ce_2O_7$ supports even it is calcinated at 900 °C. In addition, although calcinated at different temperatures, (111) facet of $La_2Ce_2O_7$ is exposed (Fig. S1).

To further investigate the catalyst, the specific surface area of the catalysts was measured. As listed in Table 2, the surface area of the catalyst first increases with calcination temperature and then decreases after achieving highest surface area at 700 °C (40.2 m^2/g). The change in surface area is consistent with the results of XRD and SEM. At calcination temperature of 800 °C, dramatic decrease in surface area is confirmed, clearly due to the sintering of support. This speculation is reinforced by the change of density as demonstrated in Table S1. At calcination temperatures above 800 °C, the density of $La_2Ce_2O_7$ is significantly higher than that of samples calcinated at temperatures between 500 and 700 °C. Although the surface area of $La_2Ce_2O_7$ is lower than 40 m^2/g, which is far lower than that of carbon materials,[14] high dispersion of Ru was still achieved as indicated in Fig. 4. It implies that the interaction between Ru and $La_2Ce_2O_7$ may be responsible for the dispersion, rather than high surface area.

Table 1
Crystal sizes and 2θ positions of $La_2Ce_2O_7$ supports with different calcination temperatures and their corresponding spent catalysts.

Sample	2θ(111)/(°)	$D_{(111)}$ (nm)	2θ(200)/(°)	$D_{(200)}$ (nm)	2θ(220)/(°)	$D_{(220)}$ (nm)	2θ(311)/(°)	$D_{(311)}$ (nm)
$La_2Ce_2O_7$-500	27.94	7.92	32.46	9.16	46.36	8.54	54.94	10.67
$La_2Ce_2O_7$-700	27.74	8.79	32.08	9.22	46.26	8.62	54.88	8.35
$La_2Ce_2O_7$-800	27.76	12.47	32.20	12.60	46.12	11.56	54.72	11.74
Spent Ru/$La_2Ce_2O_7$-500	27.58	9.80	31.96	9.91	45.92	10.67	54.62	10.80
Spent Ru/$La_2Ce_2O_7$-700	27.64	9.65	31.90	9.97	46.39	10.40	54.62	9.77
Spent Ru/$La_2Ce_2O_7$-800	27.78	12.32	32.10	12.84	46.08	12.27	54.62	12.82

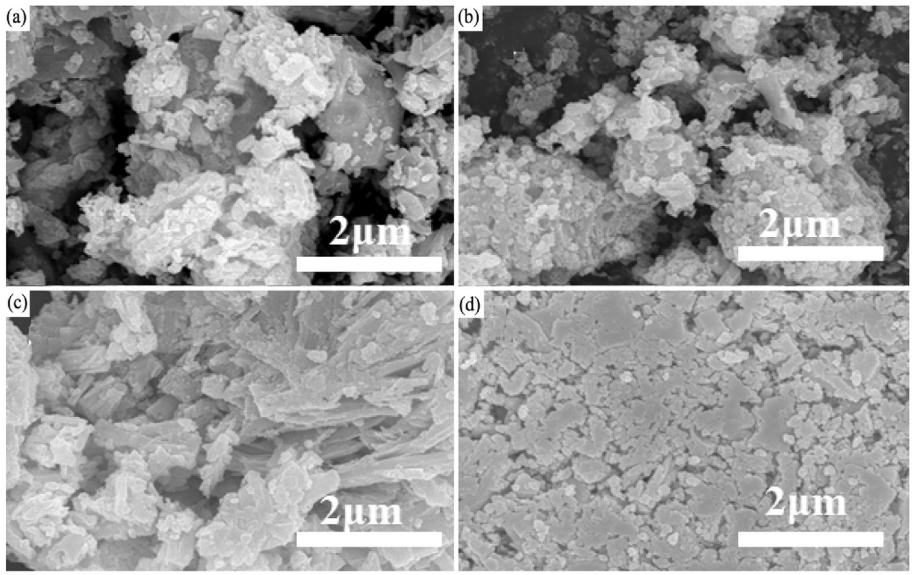

Fig. 2. SEM images of the Ru/$La_2Ce_2O_7$ catalysts. (a) Ru/$La_2Ce_2O_7$-500; (b) Ru/$La_2Ce_2O_7$-700; (c) Ru/$La_2Ce_2O_7$-800; (d) Ru/$La_2Ce_2O_7$-900.

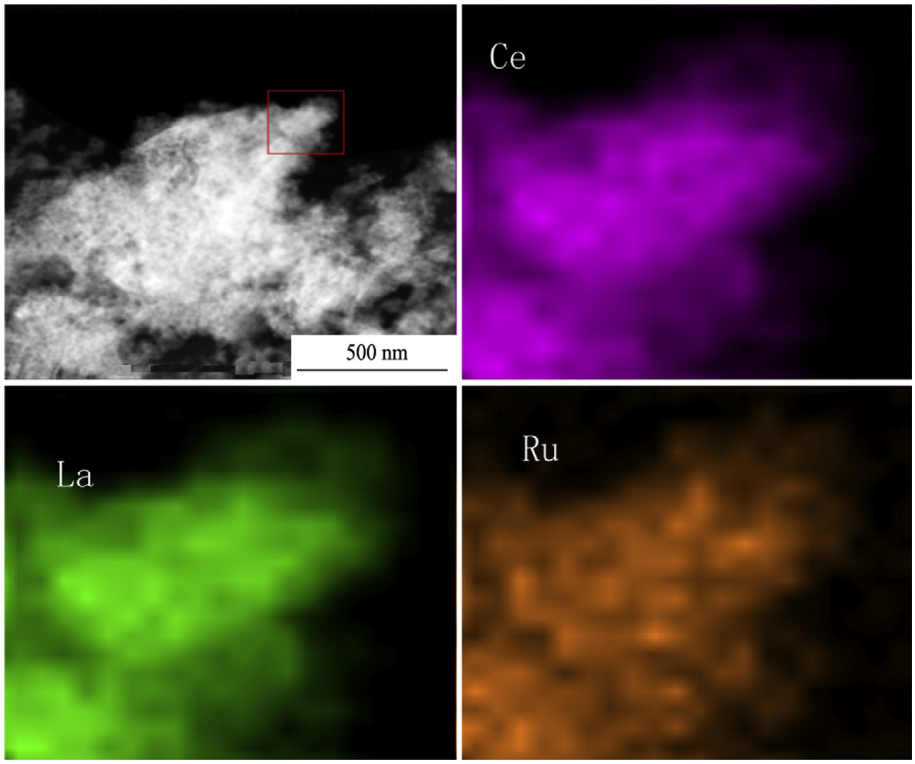

Fig. 3. EDX maps of Ru/$La_2Ce_2O_7$-800 catalyst.

3.2. Catalysis performance

The activity of $La_2Ce_2O_7$ supported Ru catalysts for ammonia synthesis was evaluated at temperatures between 375 and 450 °C, pressure of 10 MPa and GHSV of 10000 h^{-1}. As shown in Fig. 5(a) and (b), ammonia concentration of Ru/$La_2Ce_2O_7$ first increases with calcination temperature, then drops with calcination temperature significantly after achieving maximum ammonia concentration of 12.94% at calcination temperature of 800 °C (with reaction temperature of 425 °C). Compared with CeO_2 and La_2O_3 supports, the ammonia concentration of Ru/$La_2Ce_2O_7$-800 is 3.5% higher than that of Ru/CeO_2 and 9% higher than that of Ru/La_2O_7. Clearly, the formation of $La_2Ce_2O_7$ facilitates the catalytic activity for ammonia synthesis. Similar trend of reaction rate for ammonia synthesis was observed (Fig. S2).

In addition, Ru/$La_2Ce_2O_7$ catalysts exhibit high activities at low pressures. As demonstrated in Fig. 5(c), ammonia concentration of Ru/$La_2Ce_2O_7$-800 is as high as 4.7% at pressure of only 3 MPa and temperature of 375 °C and GHSV of 10000 h^{-1}. With the reaction pressure increasing to 5 MPa, ammonia concentration of 5.5% is

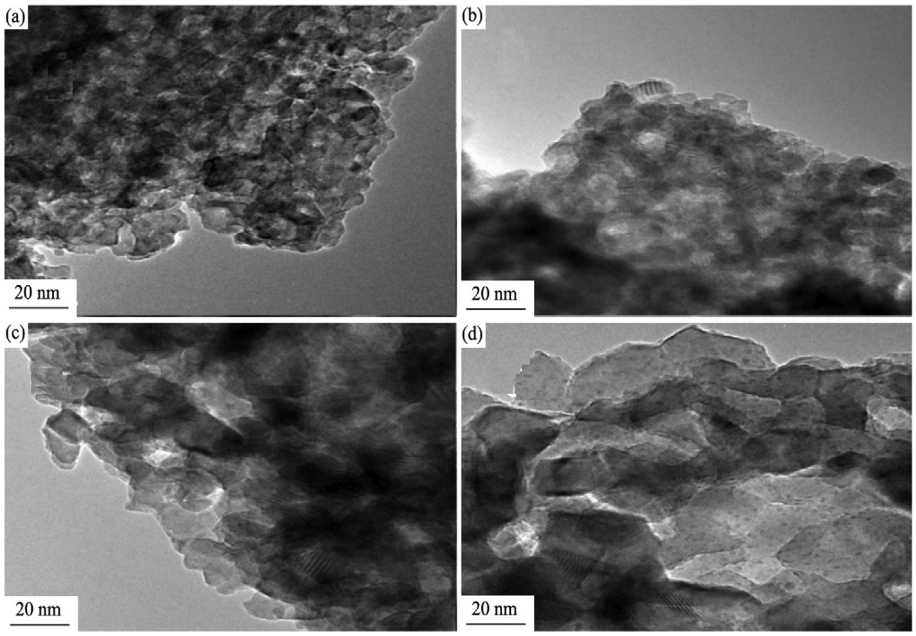

Fig. 4. TEM images of the catalysts. (a) Ru/$La_2Ce_2O_7$-500; (b) Ru/$La_2Ce_2O_7$-700; (c) Ru/$La_2Ce_2O_7$-800; (d) Ru/$La_2Ce_2O_7$-900.

Table 2
Specific surface area of Ru/$La_2Ce_2O_7$ catalysts.

Sample	S_{BET} (m^2/g)	V_{tot} (cm^3/g)	D_{de} (nm)
Ru/$La_2Ce_2O_7$-500	19.9	0.27	84.6
Ru/$La_2Ce_2O_7$-600	25.7	0.16	85.8
Ru/$La_2Ce_2O_7$-700	40.2	0.22	59.8
Ru/$La_2Ce_2O_7$-800	25.8	0.18	84.8

achieved at 375 °C and 10000 h^{-1}. Unexpectedly, Ru/$La_2Ce_2O_7$ catalyst shows much higher activity than those of similar composite catalysts reported in literature, such as Ru/$BaCeO_3$, Ru/$BaTiO_3$, Ru/$BaZrO_3$ catalysts.[12,32,33] The activity comparison of this study with the data reported in literature is listed in Table S2. As displayed in Table S2, under similar conditions, the ammonia concentration of Ru/$BaTO_3$ catalyst is 2.87% (3 MPa, 375 °C), 4.15% for Ru/$BaZrO_3$ catalyst (5 MPa and 375 °C), and 0.05% for Ru/$BaCeO_3$ (3 MPa, 400 °C). Clearly, Ru/$La_2Ce_2O_7$ catalyst possesses much higher activity.

It is worth noting that Ru/$La_2Ce_2O_7$ possesses highly thermal stability as indicated in Fig. 5(d). Following over-heating at 500 °C for 12 h, no deactivation was observed. Clearly, no significant sintering takes place even at reaction temperature of 500 °C.

4. Discussion

It has been reported that Ce^{4+} will be reduced to Ce^{3+} at high temperatures, which is helpful to donate electron to Ru atoms.[30,34] Therefore, XPS technique is adopted to verify the valence change of La, Ce and Ru. As displayed in Fig. 6, over the fresh and spent Ru/$La_2Ce_2O_7$-800 catalysts, Ce 3d, La 3d, Ru 3d spectra were recorded. According to the results, no obvious shift in binding energy of La is identified before and after the reaction. Fig. 6(a) represents the XPS spectra of the fresh and the spent Ru/$La_2Ce_2O_7$-800 catalysts for Ce 3d. It is suggested that Ce $3d_{5/2}$ and Ce $3d_{3/2}$ levels were composed of 10 various states.[35,36] The peaks at 916.7, 907.3, 901, 898.4, 888.2 and 882.5 eV are attributed to Ce^{4+} species and while peaks at 903.5, 898.8, 884.9 and 880.3 eV are assigned to Ce^{3+} species. The surface concentration of Ce^{3+} was determined by the peak volume ratio of all Ce^{3+} peaks to all Ce peaks.

This value was found to be approximately 15% and 23% for the fresh and the spent Ru/$La_2Ce_2O_7$-800, respectively. Following reaction at 475 °C, the concentration of Ce^{3+} was increased slightly. It indicates the reduction of Ce^{4+} to Ce^{3+} during ammonia synthesis reaction. It also explains the slight increase in catalytic activity after over-heating at 500 °C for 12 h (Fig. 5(d)). For the fresh Ru/$La_2Ce_2O_7$-800, two values of binding energy (E_b) of Ru 3d are detected at 284.76 and 281.77 eV, respectively, representing the typical value of Ru $3d_{3/2}$ and Ru $3d_{5/2}$.[30] Compared with the standard binding energy of zero valence Ru (280.0 eV), significant shift Ru/$La_2Ce_2O_7$-800 is readily attributed to $Ru_3(CO)_{12}$.[14] Following reaction, binding energies of Ru 3d are located at 284.74, 280.15 and 281.10 eV, respectively. Binding energy of 280.15 eV is the typical value of zero valence Ru (280.0 eV), and 284.74 eV is attributed to the Ru $3d_{5/2}$. The new peak at 281.10 eV can not be assigned to RuO_2 (280.5 and 282.1 eV) and other species of Ru. Hence, we suggest that the strong interaction between Ru and the support results in the shift of the binding energy by Ru or RuO_2.[14]

The above assumption is reinforced by the results of CO-chemisorption as listed in Table 3. Ru dispersion over $La_2Ce_2O_7$ is dramatically higher than that of La_2O_3 and CeO_2. In addition, following calcination at 700 °C, the highest surface area was achieved for $La_2Ce_2O_7$-700 (Table 2). However, it does not lead to the high dispersion of Ru (Table 3). Hence, dispersion of Ru over $La_2Ce_2O_7$ is not solely determined by surface area. According to Table 3, relatively high Ru dispersion was obtained over $La_2Ce_2O_7$-600 support, and Ru/$La_2Ce_2O_7$-800 presents higher activity (Fig. 3). Clearly, in addition to the Ru dispersion, interaction between Ru and $La_2Ce_2O_7$, especially the donation of electron to Ru plays an important role in the activity. It is noted that the Ru particle size determined by CO-chemisorption is significantly larger than that of TEM measurement. The difference may result from the stoichiometric coefficients of 1 for CO adsorption adopted in this work.

To further investigate the effect of $La_2Ce_2O_7$ on the activity of Ru catalysts, Arrhenius plots were conducted and the results are presented in Fig. 7.[37] As indicated in Fig. 7, the activation energy over La_2O_3 and CeO_2 supported catalysts is higher than that of $La_2Ce_2O_7$

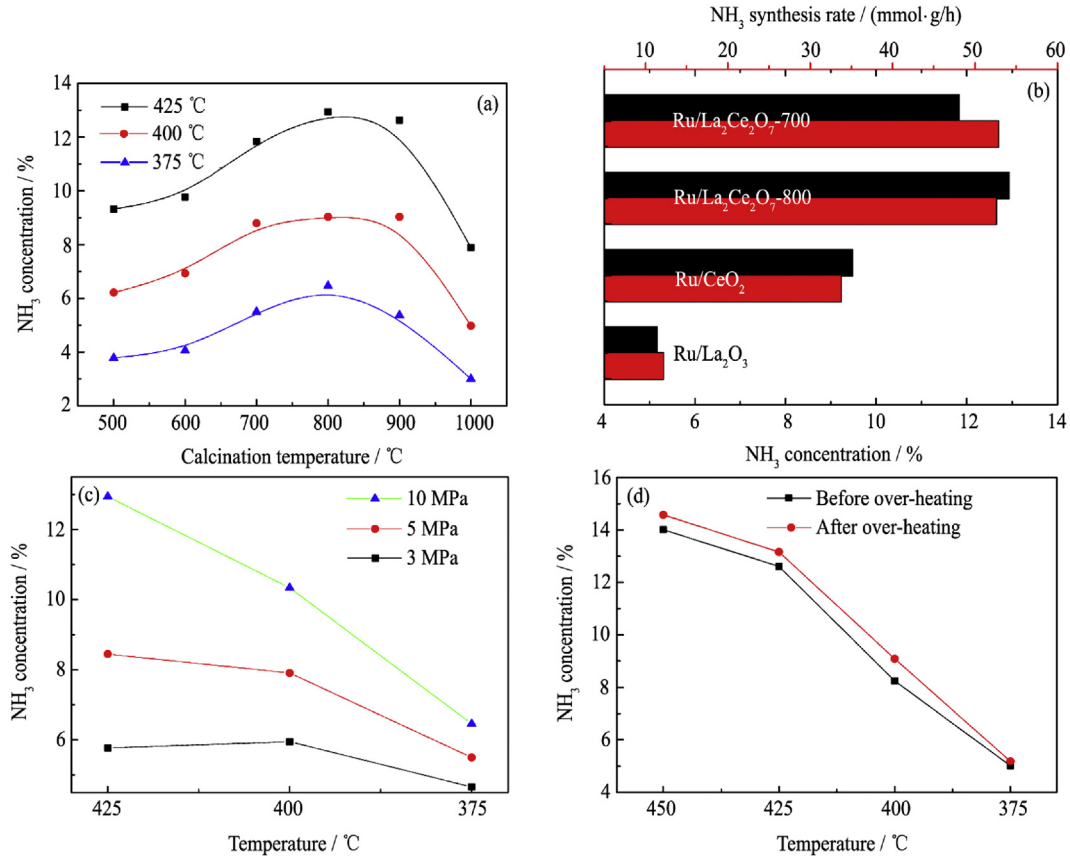

Fig. 5. Catalytic performance of $Ru/La_2Ce_2O_7$ for ammonia synthesis. (a) Effect of $La_2Ce_2O_7$ calcination temperature on the ammonia concentration (H_2/N_2 = 3:1, 10000 h^{-1} and 10 MPa); (b) Ammonia-synthesis rates and NH_3 concentration over supported Ru catalysts (H_2/N_2 = 3:1, 10000 h^{-1}, 425 °C and 10 MPa); (c) Ammonia concentration of $Ru/La_2Ce_2O_7$-800 as a function of reaction pressure (H_2/N_2 = 3:1, 10000 h^{-1}); (d) Thermal stability of $Ru/La_2Ce_2O_7$-800 (the catalyst was over heated at 5 MPa, 500 °C, 30000 h^{-1} for 12 h).

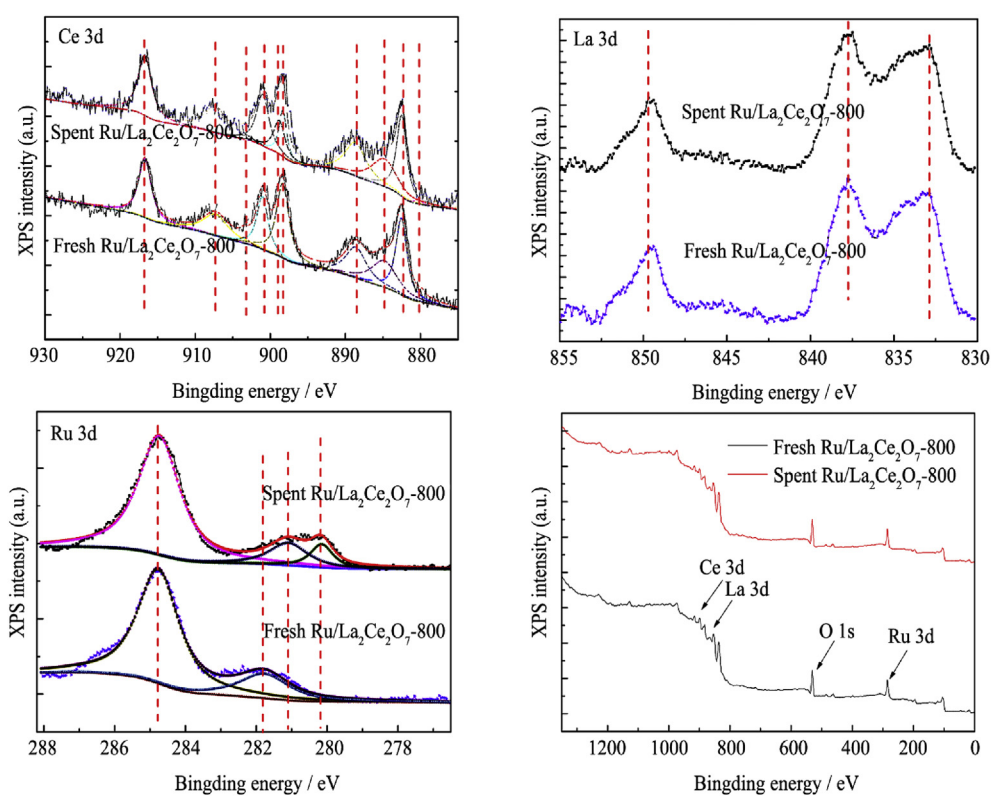

Fig. 6. XPS spectra of the fresh and spent $Ru/La_2Ce_2O_7$-800 catalyst. (a) Ce 3d spectra; (b) La 3d spectra; (c) Ru 3d spectra; (d) Ce 3d, La 3d, Ru 3d spectra of fresh and spent $Ru/La_2Ce_2O_7$-800.

Table 3
Dispersion of Ru for $Ru/La_2Ce_2O_7$ catalysts.

Sample	Monolayer CO uptake (μmol/g)	Ru dispersion%	d_{Ru} (nm)
$Ru/La_2Ce_2O_7$-500	153.5	39.0	3.4
$Ru/La_2Ce_2O_7$-600	167.6	42.6	3.2
$Ru/La_2Ce_2O_7$-700	126.3	32.1	4.1
$Ru/La_2Ce_2O_7$-800	145.7	37.1	3.6
$Ru/La_2Ce_2O_7$-900	119.0	30.3	4.4
Ru/La_2O_3	26	6.6	203.4
Ru/CeO_2	38.1	9.7	138.6

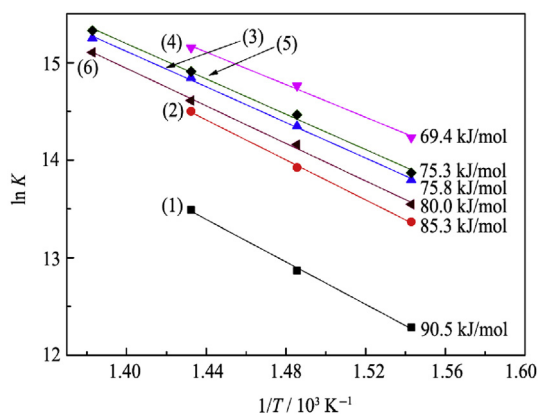

Fig. 7. Arrhenius plots over the $La_2Ce_2O_7$ supported Ru catalysts. (1) Ru/La_2O_3; (2) Ru/CeO_2; (3) $Ru/La_2Ce_2O_7$-500; (4) $Ru/La_2Ce_2O_7$-700; (5) $Ru/La_2Ce_2O_7$-800; (6) $Ru/La_2Ce_2O_7$-900.

supported catalysts. In addition, activation energy of $La_2Ce_2O_7$ supported catalysts decreases with calcination temperature above 700 °C. However, it increases significantly with calcination temperature lower than 700 °C. Clearly, $Ru/La_2Ce_2O_7$-700 catalyst achieves the lowest activation energy for ammonia synthesis (69.4 kJ/mol) with $La_2Ce_2O_7$ calcination temperature of 700 °C. This explains why $Ru/La_2Ce_2O_7$-700 possesses the highest reaction rate among all the catalysts investigated (Fig. 5(b)). The higher ammonia

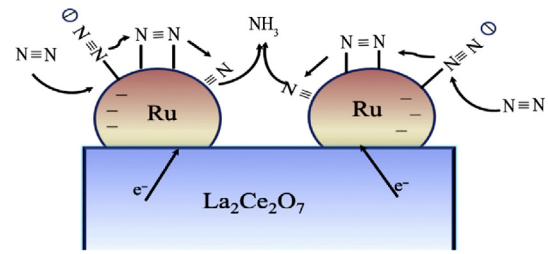

Fig. 9. Schematic illustration of electron donation from La_2Ce_2O to Ru and activation of N_2 over Ru surface.

concentration over $Ru/La_2Ce_2O_7$-800 than $Ru/La_2Ce_2O_7$-700 (Fig. 5) is attributed to the higher density (Table S1) as catalysts were evaluated based on the same volume (2 mL).

For ammonia synthesis, this enhancement of the catalytic activity is ascribed to the transfer of electrons from Ru to the anti π orbital of N_2, also known as the electron promoting effect.[38] Clearly, the donation of electrons by $La_2Ce_2O_7$ to Ru facilitates the dissociation of N_2 and hence ammonia synthesis. Therefore, a calcination of temperature above 700 °C may be necessary for the complete formation of $La_2Ce_2O_7$ crystalline. However, with calcination temperature higher than 800 °C, significant sintering was observed. Consequently, $Ru/La_2Ce_2O_7$-700 catalyst approaches the highest ammonia synthesis rate among all the catalysts studied. In addition, the small (311) facet size (see Table 1) over $La_2Ce_2O_7$-700 may also contribute to the activity which needs to be investigated in future.

As indicated in Fig. 4, no clear Ru particle is identified over $Ru/La_2Ce_2O_7$-500, $Ru/La_2Ce_2O_7$-600, $Ru/La_2Ce_2O_7$-700 and $Ru/La_2Ce_2O_7$-800. As shown in Fig. 8, the average Ru particle size is determined to be 1.5 nm for $Ru/LaCe_2O$-900 catalyst. Generally, it is accepted that ammonia synthesis over Ru is a structure sensitive reaction. Ru with B_5 structure (1.8–3.5 nm) is suitable for high activity.[39] Clearly, Ru particle size in $Ru/La_2Ce_2O_7$-500, $Ru/La_2Ce_2O_7$-600, $Ru/La_2Ce_2O_7$-700 and $Ru/La_2Ce_2O_7$-800 is even smaller than that of $Ru/La_2Ce_2O_7$-900 catalyst (1.5 nm). These Ru particle sizes are not favorable for the formation of B_5 structure.

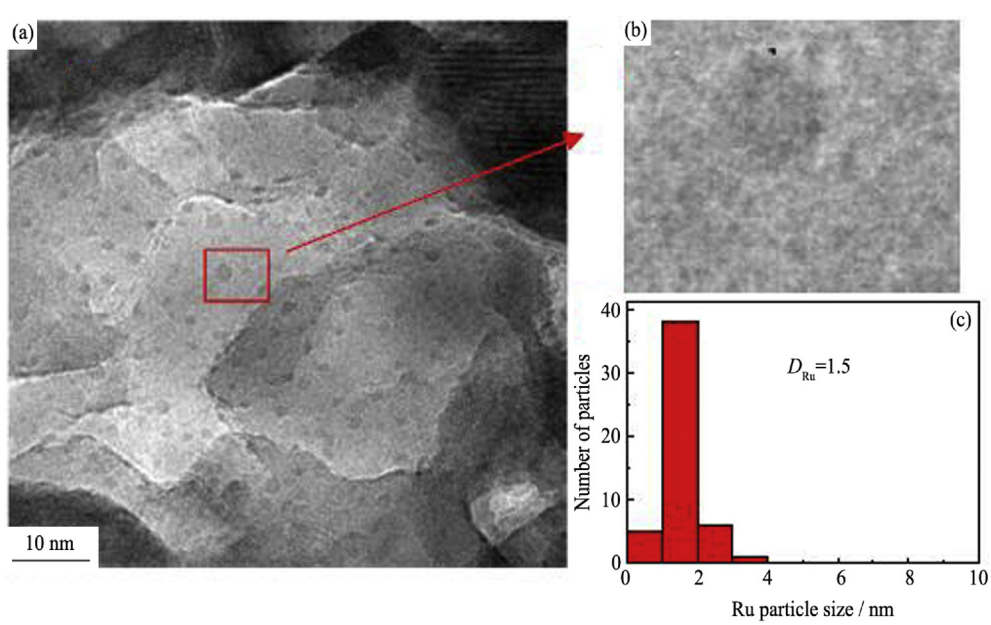

Fig. 8. TEM images for Ru/La_2Ce_2O-900 catalyst. (a) TEM image; (b) High magnitude image of Ru particle; (c) Ru particle size distributions.

Probably, as suggested by Ma et al. and Song et al., the morphology of Ru and the interaction between Ru and support play important roles in the activity of ammonia synthesis.[40,41] As discussed previously, the interaction between Ru and $La_2Ce_2O_7$ (donation of electrons from $La_2Ce_2O_7$ to Ru as proposed in Fig. 9) facilitates the improvement of catalytic activity.

5. Conclusions

$La_2Ce_2O_7$ nanoparticles were prepared by citric acid complexation method followed by calcination at varied temperatures. The formation of $La_2Ce_2O_7$ solid solution is identified following calcination at temperatures above 500 °C. Crystal sizes increase with calcination temperature elevated from 500 to 900 °C. Although with high magnitude during the measurement of TEM, no clear Ru particles were observed on $La_2Ce_2O_7$-500, $La_2Ce_2O_7$-700 and $La_2Ce_2O_7$-800. The average Ru particle size of Ru/La_2Ce_2O-900 catalyst is determined to be 1.5 nm. Compared with CeO_2 and La_2O_3 supports, the ammonia concentration of Ru/$La_2Ce_2O_7$-800 is 3.5% higher than that of Ru/CeO_2 and 9% higher than that of Ru/La_2O. In addition, Ru/$La_2Ce_2O_7$ catalyst also shows much higher activity than those of similar composite catalysts, such as Ru/$BaCeO_3$, Ru/$BaTO_3$ and Ru/$BaZrO_3$ catalysts. Following over-heating at 500 °C for 12 h, no deactivation is observed, indicating high thermal stability of Ru/$La_2Ce_2O_7$ catalysts. In addition to the high dispersion of Ru, we suggest that the strong interaction between Ru and the support leading to the donation of electrons from $La_2Ce_2O_7$ to Ru is responsible for the high activity.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jre.2018.09.010.

References

1. Chen JG, Crooks RM, Seefeldt LC, Bren KL, Bullock RM, Darensbourg MY, et al. Beyond fossil fuel-driven nitrogen transformations. *Science*. 2018;360(6391):873.
2. Mukherjee S, Devaguptapu SV, Sviripa A, Lund CRF, Wu G. Low-temperature ammonia decomposition catalysts for hydrogen generation. *Appl Catal B Environ*. 2018;226:162.
3. Qian J, An Q, Fortunelli A, Nielsen RJ, Goddard WA. Reaction mechanism and kinetics for ammonia synthesis on the Fe(111) surface. *J Am Chem Soc*. 2018;140(20):6288.
4. Sehested J, Jacobsen CJH, Tornqvist E, Rokni S, Stoltze P. Ammonia synthesis over a multipromoted iron catalyst: extended set of activity measurements, microkinetic model, and hydrogen inhibition. *J Catal*. 1999;188(1):83.
5. Florez-Orrego D, de Oliveira S. Modeling and optimization of an industrial ammonia synthesis unit: an exergy approach. *Energy*. 2017;137:234.
6. Song CF, Liu QL, Ji N, Song YJ, Kitamura Y. Conceptual design and process feasibility analysis of a novel ammonia synthesis process by efficient heat integration. *ACS Sustain Chem Eng*. 2017;5(8):7420.
7. Liu HZ. Ammonia synthesis catalyst 100 years: practice, enlightenment and challenge. *Chin J Catal*. 2014;35(10):1619.
8. Zheng XL, Wei KM. The second generation catalysis system for ammonia synthesis - ruthenium-based ammonia synthesis catalyst and its industrial application. *Prog Chem*. 2001;13(6):472.
9. Bielawa H, Hinrichsen O, Birkner A, Muhler M. The ammonia-synthesis catalyst of the next generation: barium-promoted oxide-supported ruthenium. *Angew Chem Int Edit*. 2001;40(6):1061.
10. Szmigiel D, Bielawa H, Kurtz M, Hinrichsen O, Muhler M, Rarog W, et al. The kinetics of ammonia synthesis over ruthenium-based catalysts: the role of barium and cesium. *J Catal*. 2002;205(1):205.
11. Miyazaki A, Balint L, Aika K, Nakano Y. Preparation of Ru nanoparticles supported on gamma-Al_2O_3 and its novel catalytic activity for ammonia synthesis. *J Catal*. 2001;204(2):364.
12. Wang ZQ, Lin JX, Wang R, Wei KM. Ammonia synthesis over ruthenium catalyst supported on perovskite type $BaTiO_3$. *Catal Commun*. 2013;32:11.
13. Jacobsen CJH. Boron nitride: a novel support for ruthenium-based ammonia synthesis catalysts. *J Catal*. 2001;200(1):1.
14. Han WF, Li LH, Yan HY, Tang HD, Li Z, Li Y, et al. Strong interaction of ruthenium species with graphite structure for the self-dispersion of Ru under solvent-free conditions. *ACS Sustain Chem Eng*. 2017;5(8):7195.
15. Li Y, Pan CG, Han WF, Chai HF, Liu HZ. An efficient route for the preparation of activated carbon supported ruthenium catalysts with high performance for ammonia synthesis. *Catal Today*. 2011;174(1):97.
16. Han WF, Yan HY, Tang HD, Li Y, Liu HZ. Preparation of efficient ruthenium catalysts for ammonia synthesis via high surface area graphite dispersion. *React Kinet Mech Catal*. 2014;113(2):361.
17. Zhou CH, Zhu YF, Liu HZ. Effect of samarium on methanation resistance of activated carbon supported ruthenium catalyst for ammonia synthesis. *J Rare Earths*. 2010;28(4):552.
18. Kowalczyk Z, Jodzis S, Rarog W, Zielinski J, Pielaszek J. Effect of potassium and barium on the stability of a carbon-supported ruthenium catalyst for the synthesis of ammonia. *Appl Catal A Gen*. 1998;173(2):153.
19. Zeng HS, Hihara T, Inazu K, Aika K. Effect of methanation of active carbon support on the barium-promoted ruthenium catalyst for ammonia synthesis. *Catal Lett*. 2001;76(3–4):193.
20. Chen MX, Yuan MW, Li JJ, You ZX. Ammonia synthesis over Cs- or Ba-promoted ruthenium catalyst supported on strontium niobate. *Appl Catal A Gen*. 2018;554:1.
21. Kitano M, Inoue Y, Sasase M, Kishida K, Kobayashi Y, Nishiyama K, et al. Self-organized ruthenium-barium core-shell nanoparticles on a mesoporous calcium amide matrix for efficient low-temperature ammonia synthesis. *Angew Chem Int Edit*. 2018;57(10):2648.
22. Ni J, Jing BQ, Lin JX, Lin BY, Zhao ZQ, Jiang LL. Effect of rare earth on the performance of Ru/MgAl-LDO catalysts for ammonia synthesis. *J Rare Earths*. 2018;36(2):135.
23. Niwa Y, Aika K. The effect of lanthanide oxides as a support for ruthenium catalysts in ammonia synthesis. *J Catal*. 1996;162(1):138.
24. Saito M, Itoh M, Iwamoto J, Li CY, Machida K. Synergistic effect of MgO and CeO_2 as a support for ruthenium catalysts in ammonia synthesis. *Catal Lett*. 2006;106(3–4):107.
25. Zhao J, Xu XY, Zhou W, Blakey I, Liu SM, Zhu ZH. Proton-conducting La-doped ceria-based internal reforming layer for direct methane solid oxide fuel cells. *ACS Appl Mater Interfaces*. 2017;9(39):33758.
26. Zhu ZW, Liu BT, Shen JX, Lou YX, Ji YX. $La_2Ce_2O_7$: a promising proton ceramic conductor in hydrogen economy. *J Alloys Compd*. 2016;659:232.
27. Shabanikia A, Javanbakht M, Amoli HS, Hooshyari K, Enhessari M. Effect of $La_2Ce_2O_7$ on the physicochemical properties of phosphoric acid doped polybenzimidazole nanocomposite membranes for high temperature proton exchange membrane fuel cells applications. *J Electrochem Soc*. 2014;161(14):F1403.
28. Wang JD, Xie YH, Zhang ZF, Liu RQ, Li ZH. Protonic conduction in Ca^{2+}-doped $La_2M_2O_7$ (M = Ce, Zr) with its application to ammonia synthesis electrochemically. *Mater Res Bull*. 2005;40(8):1294.
29. Murata S, Aika K. Preparation and characterization of chlorine-free ruthenium catalysts and the promoter effect in ammonia-synthesis. 2. A lanthanide oxide-promoted Ru/Al_2O_3 catalyst. *J Catal*. 1992;136(1):118.
30. Ogura Y, Sato K, Miyahara SI, Kawano Y, Toriyama T, Yamamoto T, et al. Efficient ammonia synthesis over a Ru/$La_{0.5}Ce_{0.5}O_{1.75}$ catalyst pre-reduced at high temperature. *Chem Sci*. 2018;9(8):2230.
31. Tinwala H, Shah DV, Menghani J, Pati R. Synthesis of $La_2Ce_2O_7$ nanoparticles by co-precipitation method and its characterization. *J Nanosci Nanotechnol*. 2014;14(8):6072.
32. Wang ZQ, Liu BY, Lin JX. Highly effective perovskite-type $BaZrO_3$ supported Ru catalyst for ammonia synthesis. *Appl Catal A Gen*. 2013;458:130.
33. Yang XL, Zhang WQ, Xia CG, Xiong XM, Mu XY, Hu B. Low temperature ruthenium catalyst for ammonia synthesis supported on $BaCeO_3$ nanocrystals. *Catal Commun*. 2010;11(10):867.
34. Luo XJ, Wang R, Ni J, Lin JX, Lin BY, Xu XN, et al. Effect of La_2O_3 on Ru/CeO_2-La_2O_3 catalyst for ammonia synthesis. *Catal Lett*. 2009;133(3–4):382.
35. Chang LH, Sasirekha N, Chen YW, Wang WJ. Preferential oxidation of CO in H_2 stream over Au/MnO_2-CeO_2 catalysts. *Ind Eng Chem Res*. 2006;45(14):4927.
36. Gasser-Ramirez JL, Dunn BC, Ramirez DW, Fillerup EP, Turpin GC, Shi Y, et al. A simple synthesis of catalytically active, high surface area ceria aerogels. *J Non-Crystal Solids*. 2008;354(52–54):5509.
37. Kalenczuk RJ. Cobalt promoted fused iron catalyst for ammonia synthesis. *Int J Inorg Mater*. 2000;2(2–3):233.
38. Rao CNR, Rao GR. Nature of nitrogen adsorbed on transition-metal surfaces as revealed by electron-spectroscopy and cognate techniques. *Surf Sci Rep*. 1991;13(7):221.
39. Jacobsen CJH, Dahl S, Hansen PL, Tornqvist E, Jensen L, Topsoe H, et al. Structure sensitivity of supported ruthenium catalysts for ammonia synthesis. *J Mol Catal Chem*. 2000;163(1–2):19.
40. Song Z, Cai TH, Hanson JC, Rodriguez JA, Hrbek J. Structure and reactivity of Ru nanoparticles supported on modified graphite surfaces: a study of the model catalysts for ammonia synthesis. *J Am Chem Soc*. 2004;126(27):8576.
41. Ma ZW, Zhao SL, Pei XP, Xiong XM, Hu B. New insights into the support morphology-dependent ammonia synthesis activity of Ru/CeO_2 catalysts. *Catal Sci Technol*. 2017;7(1):191.